

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Semester III Examination, 2016-17

**Paper No. MIB-301: Algebra (Old Course)**

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

**Question 1.** (Compulsory — 2 marks each.)

- (a) Give an example of each of the following matrices: symmetric, skew-symmetric, Hermitian, and skew-Hermitian.
- (b) If  $A$  and  $B$  are Hermitian matrices, show that  $AB + BA$  is Hermitian and  $AB - BA$  is skew-Hermitian.
- (c) Define the Echelon form of a matrix. What is the rank of a  $4 \times 4$  matrix whose entries are all 1?
- (d) Let  $*$  be the binary operation defined on the set  $\mathbb{Q}_+$  of all positive rational numbers by  $a * b = \frac{ab}{5}$ . Find the identity element and the inverse of an element  $a \in \mathbb{Q}_+$ .
- (e) Show that in a group  $G$ , if  $(ab)^2 = a^2b^2$  for all  $a, b \in G$ , then  $G$  must be abelian.
- (f) Let  $G$  be a group. If  $a \in G$  is of order  $n$ , show that  $o(g^{-1}ag) = n$  for any  $g \in G$ .
- (g) If  $H$  is a subgroup of a group  $G$ , show that  $HH = H$ .
- (h) Let  $G = \langle a \rangle$  be a cyclic group of order 13. Find the orders of  $a^5$  and  $a^9$ .
- (i) Define cyclic permutation, and even and odd permutations, and give one example of each.
- (j) Show that a group of prime order is cyclic.
- (k) Let  $N$  be a normal subgroup of  $G$  and  $K$  be a subgroup of  $G$ . Show that  $N \cap K$  is a normal subgroup of  $K$ .

**Question 2.**

- (a) Show that every square matrix can be expressed uniquely as  $P + iQ$ , where  $P$  and  $Q$  are Hermitian matrices. [6]
- (b) Let  $A$  be a square matrix of order  $n$ . Show that:

$$A(\text{adj } A) = (\text{adj } A)A = |A|I_n$$

where  $I_n$  is the identity matrix of order  $n$ .

[6]

**Question 3.**

- (a) By using elementary transformations, find the rank of the following matrix:

$$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

[6]

- (b) Find all solutions of the following system of linear equations:

$$3x + 4y - z - 6w = 0$$

$$2x + 3y + 2z - 3w = 0$$

$$2x + y - 14z - 9w = 0$$

[6]

**Question 4.**

- (a) Prove that a finite semigroup  $G$  in which both the cancellation laws hold is a group. Give an example in which this result fails to hold. [6]
- (b) Prove that the union of two subgroups of a group  $G$  is a subgroup of  $G$  if and only if one is contained in the other. [6]

**Question 5.**

- (a) Express the permutation:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 3 & 2 & 6 & 7 & 1 & 5 \end{pmatrix}$$

as the product of disjoint cycles, determine whether it is even or odd, and find its order. [6]

- (b) Let  $f : G \rightarrow G'$  be a group homomorphism. Show that  $f$  is a monomorphism (one-to-one homomorphism) if and only if  $\ker f = \{e\}$ , where  $e$  is the identity in  $G$ . [6]

**Question 6.**

- (a) Prove that the order of a cyclic group  $G = \langle a \rangle$  is equal to the order of its generator  $a$ . [6]
- (b) Prove that a subgroup  $H$  of a group  $G$  is a normal subgroup of  $G$  if and only if  $g^{-1}hg \in H$  for each  $g \in G, h \in H$ . [6]

**Question 7.**

- (a) Let  $f : G \rightarrow G'$  be an onto group homomorphism. If  $H$  and  $H'$  are normal subgroups of  $G$  and  $G'$  respectively, show that  $f(H)$  and  $f^{-1}(H')$  are normal subgroups of  $G'$  and  $G$  respectively. [6]
- (b) If  $H$  and  $K$  are two subgroups of a group  $G$  such that  $H$  is normal in  $G$ , prove that:

$$(HK)/H \cong K/(H \cap K)$$

[6]